Roll No.....

MID SEMESTER EXAMINATION 2025

Name of the Program: **M.Tech** Semester: **First**

Name of the Course: **(Electric Vehicle Technology)** Course Code: **MEV 101**

Time: **90 Minutes** Maximum Marks: **50**

Note:

(i) Answer **all the questions** by choosing *any one of the sub questions.*

(ii) Each question carries 10 marks.

Q1	(10 marks)	
(a)	Use Jacobi iteration method to solve the system of linear equations: $$2x - 3y + z = -1$$ $$x + 4y + 5z = 25$$ $$3x - 4y + z = 2$$	**CO1**
	OR	
(b)	What is Newton Raphson's method? Find a real root of $x - \cos x = 0$ correct to three decimal places.	
Q2	(10 marks)	
(a)	Find real root of equation $2x - \log_{10} x = 7$ correct to four decimal places using Bisection method.	**CO1**
	OR	
(b)	Prove that if f(x) be a polynomial of degree n in x, then the nth difference of f(x) is constant and the $(n+1)^{th}$ and higher differences are zero.	
Q3	(10 marks)	
a)	Solve the system of equations using the Decomposition method. $$x_1 + x_2 - x_3 = 2$$ $$2x_1 + 3x_2 + 5x_3 = -3$$ $$3x_1 + 2x_2 - 3x_3 = 6$$	**CO2**
	OR	
b)	Solve the polynomial equation by Birge- Vieta method. $$x^4 + 7x^3 + 24x^2 - 15 = 0$$	**CO2**
Q4	(10 marks)	
a)	Define Cubic Splines. Find the cubic spline corresponding to the interval (0,1) and (1,2) from the following table:	**CO2** **CO1**

x:	0	1	2	3
y:	1	2	33	244

	OR	

b) Apply Hermite's formula to interpolate for sin (1.05) from the following data:

X:	Sinx	Cosx
1.00	0.84147	0.54030
1.10	0.89121	0.45360

Q5 Cramer rule.(10 marks)

a) Solve the system of equations by Relaxation method.

$$10x_1 - 2x_2 - 2x_3 = 6$$
$$-x_1 + 10x_2 - 2x_3 = 7$$
$$-x_1 - x_2 + 10x_3 = 8$$

CO2, CO1

OR

(b) Define rate of convergence. Prove that Rate of convergence of Newton Raphson's method is quadratic.